Foreign Exchange Trade Journal and Risk Control System

ADVENTIST UNIVERSITY OF CENTRAL AFRICA (AUCA)

INSY 8311 Database Development with PL/SQL

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Problem Statement

Most retail foreign exchange (forex) traders lose money. The European Securities and Markets Authority found that 74–89% of retail CFD accounts lose money, with average losses between €1,600 and €29,000 per client (European Securities and Markets Authority [ESMA], 2018). Ben-David et al. (2018) studied 3,103 retail traders and found they trade larger and more often after winning weeks, while past results did not predict future success. Heimer and Simsek (2019) showed that capping leverage cut trading volume and the losses of high-leverage traders. Brokers' tools, such as tastyfx's Trade Analytics (n.d.), report results after the fact and do not block a trade that breaks the trader's own limits. Traders therefore lack a personal system that records every trade and enforces risk limits at the moment of trading.

Proposed Solution

I will build an Oracle PL/SQL database that works as a trade journal and risk control system. Each trade is stored with its entry, exit, volume and result, and is converted to the account currency using the European Central Bank (ECB) rate of the trade date. PL/SQL functions calculate win rate, profit factor, drawdown and risk per trade. Triggers reject trades that break the risk rules stored in a RISK_RULES table, and an audit log records every attempt, accepted or blocked.

Data Source

Trades come from the public Kaggle dataset FX_Daracoin_Strategy_Tester (Malek Mohammadi, 2024, MIT licence): 662 EURUSD deals from 26 January to 26 December 2023, which pair into 331 trades (142 wins, 188 losses, 1 break-even) on an account that grows from 50,000 to 194,826.96. Exchange rates come from the ECB euro foreign exchange reference rates (ECB, n.d.), published once per working day and used with the ECB's terms of use. Both are public and contain no personal data.

Planned Tables

The design follows third normal form (3NF) with eight tables:

Table	Purpose
TRADERS	Trader profile and experience level
ACCOUNTS	Trading accounts, currency and starting balance
CURRENCY_PAIRS	Pair code, base and quote currencies, contract size
EXCHANGE_RATES	ECB daily rate per currency and date
STRATEGIES	Named strategies used for tagging trades
TRADES	Entry/exit time and price, volume, stop-loss, profit
RISK_RULES	Per-trade and daily loss limits per account
AUDIT_LOG	Every trade attempt with result and reason

Planned PL/SQL Features

- **Functions:** calculate profit from price difference × volume × contract size (100,000), win rate, profit factor and risk per trade as a percentage of balance.
- **Procedures:** load trades and ECB rates, close a trade, and generate a performance summary per account.
- **Cursors:** loop over trades in date order to compute running balance and maximum drawdown (14.9% in the sample data).
- **Triggers:** a BEFORE INSERT trigger on TRADES blocks any trade risking more than 2% of the balance; a second rule blocks new trades once the daily loss passes 3% (this flags 10 historical days); another trigger writes every attempt to AUDIT_LOG.
- **Package:** FX_TRADING_PKG groups the functions and procedures above.
- **Exception handling:** named exceptions for breached risk limits, missing exchange rates and invalid data.

Innovation

The innovation is an advanced analytics dashboard built in Power BI and connected live to the Oracle database. It shows the equity curve, win rate, drawdown, risk per trade and the trades blocked by the risk rules. The business rules stay in PL/SQL; Power BI only displays the results.

References

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